Architecture Rationale

Pharmacy Inventory Management System | Lab 3 | Submission set B

Architecture selection: Layered architecture

The system is structured into presentation, application/business, and data layers. The Staff Dashboard and Supplier Portal provide the user-facing entry points. Core pharmacy functions—including Inventory & Batch Management, FEFO Stock Allocation, Expiry Monitoring & Alerts, Purchase Order Management, and Reports & Performance Benchmarking—sit in the application/business layer. The Pharmacy Database forms the persistence layer for the records used by these functions.

Why this structure fits the pharmacy workflow

1. Reliable batch selection. Stock quantity and expiry information must be considered together when items are dispensed. Locating batch validation and FEFO selection in the business layer keeps this decision in one place: the system can identify eligible batches and prioritize the one with the earliest expiry, rather than allowing each screen to implement its own selection rule.

2. A clearer replenishment path. Expiry and stock-level checks can raise a replenishment requirement for Purchase Order Management. Supplier-facing order access is kept distinct from the pharmacy's internal inventory workflow. This separation supports the purchase-order process while avoiding unnecessary exposure of internal stock operations.

Security consideration

The presentation layer does not directly own inventory or purchase-order rules. Requests pass through the business layer, where role-based checks can be applied. A supplier account can therefore be restricted to dispatched purchase orders associated with that supplier, while staff-only activities—such as maintaining batches, viewing internal stock, and carrying out dispensing-related operations—remain within protected business functions.

Performance consideration

Batch selection can be handled by the FEFO and inventory services using indexed batch and expiry-date information stored in the database. Keeping these operations separate from presentation tasks avoids placing stock-selection logic in the user interface and provides a focused path for inventory queries. This arrangement is consistent with the performance and availability goals identified for the system in Lab 1.

Component and interface coverage

The component view includes the Staff Dashboard, Supplier Portal, Inventory & Batch Management, FEFO Stock Allocation, Expiry Monitoring & Alerts, Purchase Order Management, Pharmacy Database, and Reports & Performance Benchmarking. Its interfaces represent stock requests, eligible-batch lookup, expiry data, replenishment triggers, supplier access to dispatched orders, database access, and reporting data. Provided and required interfaces are shown separately, with connectors indicating interaction direction.